
JavaScript Tutorial

Ref: <https://www.w3schools.com/> and
other internet based tutorials

JavaScript

- JavaScript is a client-side scripting language that runs entirely inside the web browser.
 - JavaScript brings a dynamic functionality to our websites.
 - JavaScript is one of the 3 languages all web developers must learn:
 1. HTML to define the content of web pages
 2. CSS to specify the layout of web pages
 3. JavaScript to program the behavior of web pages
 - Web pages are not the only place where JavaScript is used. Many desktop and server programs use JavaScript. Some databases, like MongoDB and CouchDB, also use JavaScript as their programming language.
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Functionality of JavaScript

- JavaScript Can Change HTML Content

- One of many JavaScript HTML methods is getElementById().

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>What Can JavaScript Do?</h2>
```

```
<p id="demo">JavaScript can change HTML content.</p>
```

```
<button type="button"
```

```
onclick='document.getElementById("demo").innerHTML = "Hello  
JavaScript!'">Click Me!</button>
```

```
</body>
```

```
</html>
```

- JavaScript Can Change HTML Attribute Values

```

```

```
<button
```

```
onclick="document.getElementById('myImage').src='pic_bulbon.gif'">Turn on the light
```

```
</button>
```

```
<button
```

```
onclick="document.getElementById('myImage').src='pic_bulboff.gif'">Turn off the light
```

```
</button>
```

■ JavaScript Can Change HTML Styles (CSS)

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>What Can JavaScript Do?</h2>
```

```
<p id="demo">JavaScript can change the style of an HTML  
element.</p>
```

```
<button type="button"  
onclick="document.getElementById('demo').style.fontSize='35  
px'">Click Me!</button>
```

```
</body>
```

```
</html>
```

- JavaScript Can Hide HTML Elements

- ❑ Hiding HTML elements can be done by changing the display style
- ❑ Example
- ❑ *`document.getElementById("demo").style.display = "none";`*

- JavaScript Can Show HTML Elements

- ❑ Showing hidden HTML elements can also be done by changing the display style:
 - ❑ Example
 - ❑ *`document.getElementById("demo").style.display = "block";`*
-

Including JavaScript

- In HTML, JavaScript code must be inserted between `<script>` and `</script>` tags.

- Example

```
<script>
```

```
document.getElementById("demo").innerHTML = "My First  
JavaScript";
```

```
</script>
```

External JavaScript

- Scripts can also be placed in external files:

- External file: myScript.js

```
function myFunction() {  
    document.getElementById("demo").innerHTML = "Paragraph  
changed.";} 
```

- JavaScript files have the file extension .js.
- To use an external script, put the name of the script file in the src (source) attribute of a <script> tag, like:

```
<script src="myScript.js"> </script>
```

- You can place an external script reference in <head> or <body> as you like.
 - External scripts cannot contain <script> tags.
-

External JavaScript

- External scripts can be referenced with a full URL or with a path relative to the current web page.

- Example: full URL to link to a script

```
<script src="https://www.w3schools.com/js/myScript1.js">  
</script>
```

- Example: script located in a specified folder on the current web site

```
<script src="/js/myScript1.js"> </script>
```

- Example: script located in the same folder

```
<script src="myScript1.js"> </script>
```

JavaScript Output

- JavaScript can "display" data in different ways:
 - Writing into an HTML element, using innerHTML.
 - Writing into the HTML output using document.write().
 - Writing into an alert box, using window.alert().

- Example 1: using innerHTML

```
<p>My First Paragraph.</p>
```

```
<p id="demo"></p>
```

```
<script>
```

```
document.getElementById("demo").innerHTML = 5 + 6;
```

```
</script>
```

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- Example 1: using innerHTML

```
<p>My First Paragraph.</p>
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```

```
<script>
```

```
document.getElementById("demo").innerHTML = 5 + 6;
```

```
</script>
```

JavaScript Output

- Example 2: Using document.write()
- The document.write() method should only be used for testing.

<p>My first paragraph.</p>

<script>

document.write(5 + 6);

</script>

- Example 3: Using window.alert()
- We can use an alert box to display data:

<script>

window.alert(5 + 6);

</script>